

Learning target 1.6, version 1

Explain the relationships between a function, its derivative, and its second derivative. Interpret information about these objects in real-world contexts.

Students in a recent calculus class were playing a Mad Libs game about the relationships between shapes of functions and values of the derivative:

Whenever the _____ is _____, the _____ is _____.
 noun adjective noun adjective

For each of the sentences they generated below, explain why it is correct or incorrect.
(Hint: I don't think it's possible to generate a good explanation without saying the word "slope.")

- (a) Whenever the original function is decreasing, the derivative is negative.
- (b) Whenever the derivative is increasing, the original function is positive.
- (c) Whenever the second derivative is positive, the original function is concave up.
- (d) Whenever the derivative is increasing, the second derivative is positive.

Learning target 1.6, version 2

Explain the relationships between a function, its derivative, and its second derivative. Interpret information about these objects in real-world contexts.

Students in a recent calculus class were playing a Mad Libs game about the relationships between shapes of functions and values of the derivative:

Whenever the _____ is _____, the _____ is _____.
 noun adjective noun adjective

Unfortunately, class time ended before they finished a few of their sentences. Help them fill in the blanks correctly, and explain why your choices are correct.

(Hint: I don't think it's possible to generate a good explanation without saying the word "slope.")

(a) Whenever the derivative is _____, the second derivative is negative.

(b) Whenever the original function is concave up, the derivative is _____.

(c) Whenever the _____ is positive, the _____ is increasing.

(d) Whenever the _____ is negative, the _____ is _____.
(There are several correct fill-ins!)